# HOMEWORK 4

## Exercise 1: Custom Module (100 points)

Create a Python module named utils.py containing the following functions:

### (a) Convert Numerical Grades to Letter Grades

Write a function letter_grade() that returns the letter grade corresponding to a numerical grade. The function should take two inputs: num_grade (a float) and table (a list of tuples containing grade ranges and letter equivalents).

**CODE:**

```
import math
#Our new module, utils.py
#We use to table to consolidate all letter grades and to number grade range.
table = [(90,100,"A"),(87,89,"A-"),(83,86,"B+"),
(80,82,"B"),(77,79,"B-"),(73,76,"C+"),
(70,72,"C"),(67,69,"C-"),(63,66,"D+"),
(60,62,"D"),(51,59,"D-"),(0,50,"F")]
#We now create our function to equate number grades to letter grades.
def letter_grade(score):
    score = math.floor(score)
#Now we have to check if the score provided matches any list within the tuple for each tuple in the list
    for grade in table:
      #Since for each loop its a specific tuple, for this tuple, we have the max in min indexed, and the letter grade indexed.
        if score >= grade[0] and score <= grade[1]:
            return grade[2]
    return "N/A"

if __name__ == "__main__":
  score = float(input("Enter your score: "))
  grade = letter_grade(score)
  print(grade)
```

```
Enter your score: 89.5
A-
```

### (b) Student Loan Payment Calculator

Write a function student_loan_payment() that calculates the monthly payment for a student loan using the math expression:
where:

```
M = Monthly payment
P = Loan principal (initial borrowed amount)
r = Monthly interest rate (annual rate divided by 12, as a decimal)
n = Total number of payments (loan term in years × 12)
```

Example:

```
print(utils.student_loan_payment(30000, 0.05, 10))
```

```
print(utils.student_loan_payment(50000, 0.04, 15))
print(utils.student_loan_payment(10000, 0.0, 5))
```

Output:

```
318.2
369.84
166.67
```

## Code:

```
# Student Loan Payment Finder
import math
def student_loan_payment(loan_amount, interest_rate, loan_term):
# Specific Equations for different rates, so we use if-else to represent the different scenarios  if interest_rate > 0:
    # annual rate to monthly and years to months
    monthly_interest_rate = interest_rate / 12
    total_payments = loan_term * 12
    # calucakte monthly payment
    if interest_rate > 0:
      monthly_payment = (loan_amount * monthly_interest_rate) / (1 - (1 + monthly_interest_rate)**(-total_payments))
    elif interest_rate == 0:
      total_payments = loan_term * 12
      monthly_payment = loan_amount / total_payments
    else:
      return "Interest rate cannot be negative, try again."
    return round(monthly_payment, 2)

if __name__ == "__main__":
  # Example Usage for Student Loan Payment Calculator
  loan_amount = float(input("Enter the loan amount: "))
  interest_rate= float(input("Enter the annual interest rate as a decimal: "))
  loan_term_in_years = float(input("Enter the loan term in years: "))
  print(student_loan_payment(loan_amount, interest_rate, loan_term_in_years))
```

```
Enter the loan amount: 30000
Enter the annual interest rate as a decimal: 0.05
Enter the loan term in years: 10
318.2
```

# (c) Word Count for Essay Writing

Write a function that counts the number of words and characters in a given text.

```
Example:
essay = "Python is a powerful programming language that is widely used in
various fields."
words, characters = utils.word_count(essay)
print(f"The text has {words} words and {characters} characters")
```

Output:

```
The text has 13 words and 68 characters
```

## Code

```
#Word and Character Count Finder
def word_count(essay):
  #The .split makes each thing separated by whitespace its own word in a list, and len of that list technically provides the wordcount
  word_count = len(essay.split())
  #The len provides number of characters. Replace whitespace with nothing so we can have whitespace excluded
  character_count = len(essay.replace(" ", ""))
  return word_count, character_count

if __name__ == "__main__":
  # Get user input for the essay
  essay = input("Enter your essay text: ")

  # Calculate word and character counts
  words, characters = word_count(essay)

  # Print the results
  print(f"The text has {words} words and {characters} characters")
```

```
Enter your essay text: Python is a powerful programming language that is widely used in various fields.
The text has 13 words and 68 characters
```

# (d) Temperature Converter

Write a function that converts Celsius to Fahrenheit and Fahrenheit to Celsius, based on the unit provided.

```
Example:
print(utils.convert_temperature(0, "C"))
print(utils.convert_temperature(100, "C"))
print(utils.convert_temperature(32, "F"))
print(utils.convert_temperature(212, "F"))
print(utils.convert_temperature(50, "X"))
```

Output:

```
32.0
212.0
0.0
100.0
Invalid unit! Use 'C' for Celsius or 'F' for Fahrenheit.
```

## Code:

```
#Temperature Converter
def Temperature_Converter(temperature, variable):
  #I chose this method of if else with varaiables becauase it tells us what we're working with and what we need to convert with what equations
  if variable.upper() == 'C':
    # Convert Celsius to Fahrenheit
    Temperature = (temperature * (9/5)) + 32
    return f"{Temperature:.1f} F"
  elif variable.upper() == 'F':
    # Convert Fahrenheit to Celsius
    Temperature = (temperature - 32) * (5/9)
    return f"{Temperature:.1f} C"
  else:
    return "Invalid unit! Use 'C' for Celsius or 'F' for Fahrenheit"

if __name__ == "__main__":
  # Get user input
    temperature = float(input("Enter the temperature: "))
    variable = input("Enter the unit ('C' for Celsius, 'F' for Fahrenheit): ")
    converted_temperature = Temperature_Converter(temperature, variable)
    print(converted_temperature)
```

```
Enter the temperature: 32.0
Enter the unit ('C' for Celsius, 'F' for Fahrenheit): F
0.0 C
```

# (e) Display Module Functions in a Program

```
Write a Python script that imports utils.py and demonstrates the usage of
all implemented
functions.
```

```
import utilities
print(utilities.letter_grade(69))
print(utilities.student_loan_payment(30000,0.05,10))
print(utilities.word_count("Python is a powerful programming language that is widely used in various fields."))
print(utilities.Temperature_Converter(32, "F"))
```

```
C-
318.2
(13, 68)
0.0 C
```

# Exercise 2: Bonus (10 points)

## (a) Create a Custom Function: Add a function of your own to utils.py that performs a useful task.

```python
import math

def Solve_Permutations(Number_of_options,Number_to_be_chosen):
  if Number_of_options >= Number_to_be_chosen:
    Permutation = math.factorial(int(Number_of_options)) / math.factorial(int(Number_of_options - Number_to_be_chosen))
    return Permutation
  else:
    return "Number of options must be greater than or equal to the number to be chosen"

if __name__ == "__main__":
  Number_of_options = int(input("Enter the total number of options: "))
  Number_to_be_chosen = int(input("Enter the number to be chosen: "))

  permutation_result = Solve_Permutations(Number_of_options, Number_to_be_chosen)
  print(permutation_result)
```

```
Enter the total number of options: 9
Enter the number to be chosen: 6
60480.0
```

## (b) Share Your Function: Post your function code and a short description of how it works in the class Discussion Forum on Canvas and, also here in this report! Please try not to repeat functions that others have already proposed.

This is a permutation! Using the mathematical formulas I have learned for partial permutations, I applied any restrictions (for example, the number of options being greater than the number chosen. Next I allowed user input to find the permutations.